

HorizonLink-SMD

argos-smd

The Argos SMD module is a small PCB (2x2cm) dedicated to act as a standalone or a coprocessor for any new design which required a satelite communication. The module is based on STM32WL55 processor and combined with GR5504 power amplifier allowing a comunication up to 27dBm. This module which pass the CLS certification is ready to be used.

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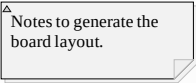
- Sheet 1: Project overview (this page)
- Sheet 2: module-BGA-RG
- Device Sheets:
 - Sheet 3: stm32wl55jci7-details
 - Sheet 4: STM32RF_power_SMPS
 - Sheet 5: CLS_1W_3.3V_Lineup

argos-smd-module



Legend

- General comment such as function title, configuration, ...
- Text to be added to silkscreen.
- Warning text.



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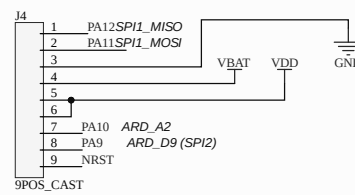
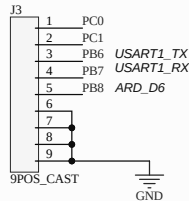
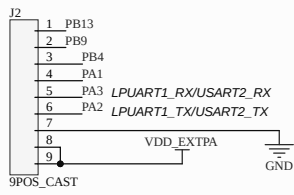
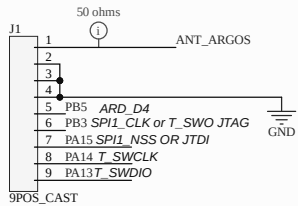
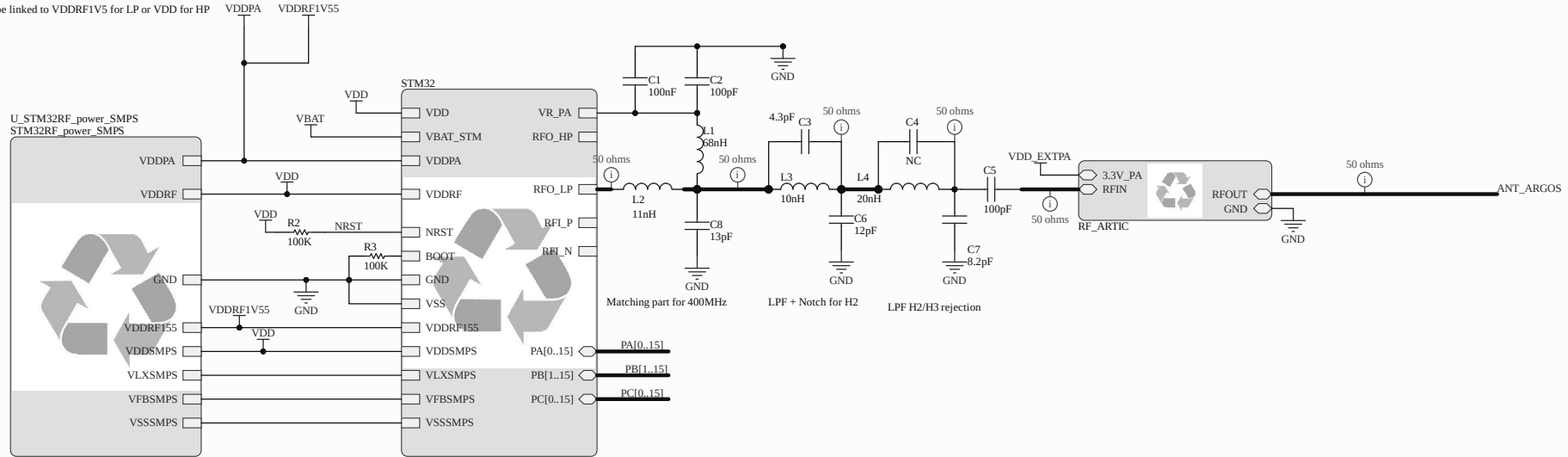
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Title HorizonLink-SMD			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org UK - Portsmouth	
Size: A4	Reference: argos-smd	Revision: 1.0		
Date: 22/11/2024	Time: 16:59:52	Sheet 1 of 5		
File: D:\Altium\Projects\stm32wl-dev\Projects\argos-smd-hw\Project_Overview.SchDoc				

VDD = 1.8 to 3.6V : External supply for system
VDDSMPS = 1.8V to 3.6V : External supply for SMPS Step down converter
VFBSMPS = 1.45V to 1.62V (1.55v) : External supply main system regulator
VDDA = 0 to 3.6V : 1.71V DAC min, 1.8V buffer, ADC 1.62V VREFBUF = 2.24V
VDDRF = 1.8 to 3.6V :external power supply for radio (VDD)
VDDRF1V = 1.45V to 1.62V for radio
VBAT = 1.55V to 3.6V : RTC Tamp, external clock...
VREF-VREF+: input ref voltage for ADC

VDDPA should be linked to VDDRF1V5 for LP or VDD for HP



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Size: A3	Reference: argos-smd	Revision: 1.0			
Date: 22/11/2024	Time: 16:59:52	Sheet2 of 5			
File: D:\Altium\Projects\stm32wl-dev\Projects\argos-smd-hw\module-BGA-RG.SchDoc					

A

B

C

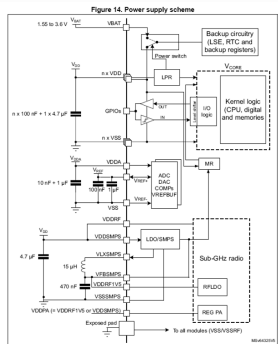
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A

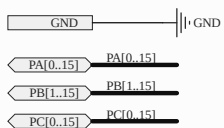
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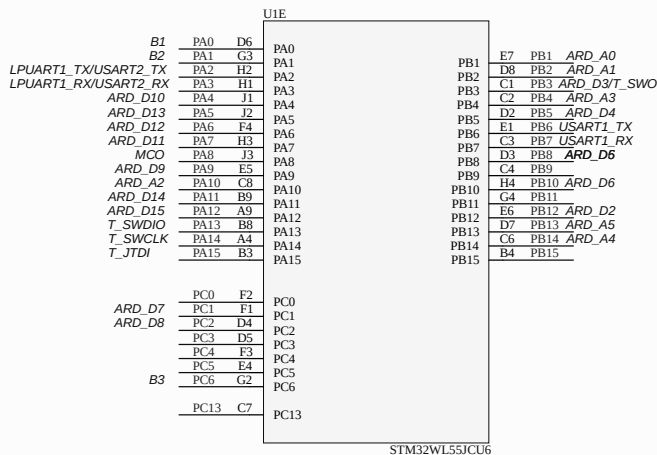
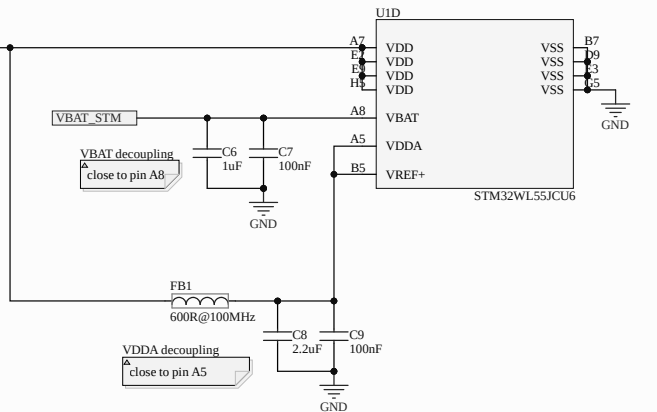
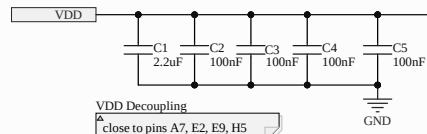
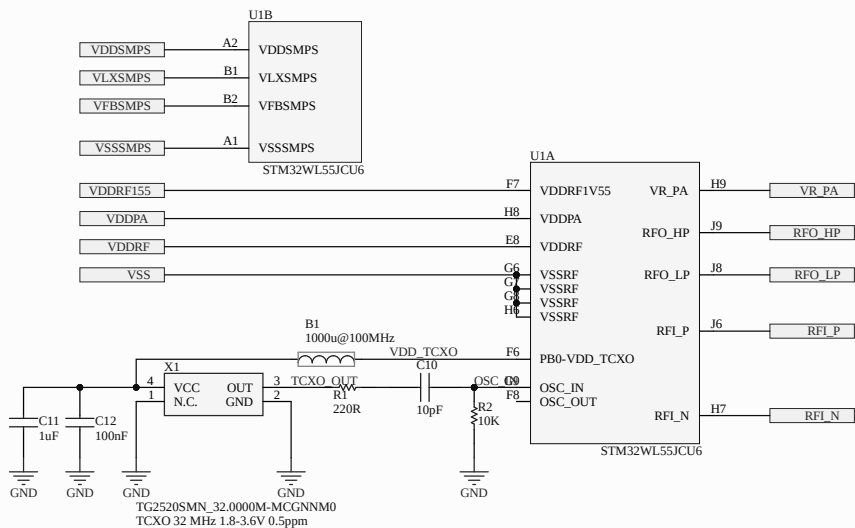
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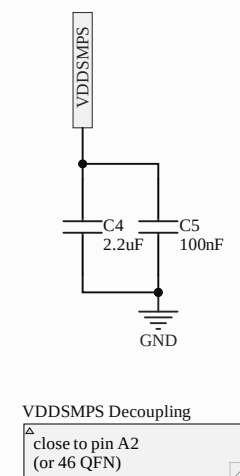


Caution: Each power supply pair (such as V_{DD}/V_{SS} or V_{DD}/V_{SS}) must be decoupled with filtering ceramic capacitors as shown in the above figure. These capacitors must be placed as close as possible to (or below) the appropriate pins on the underside of the PCB to ensure the good functionality of the device.

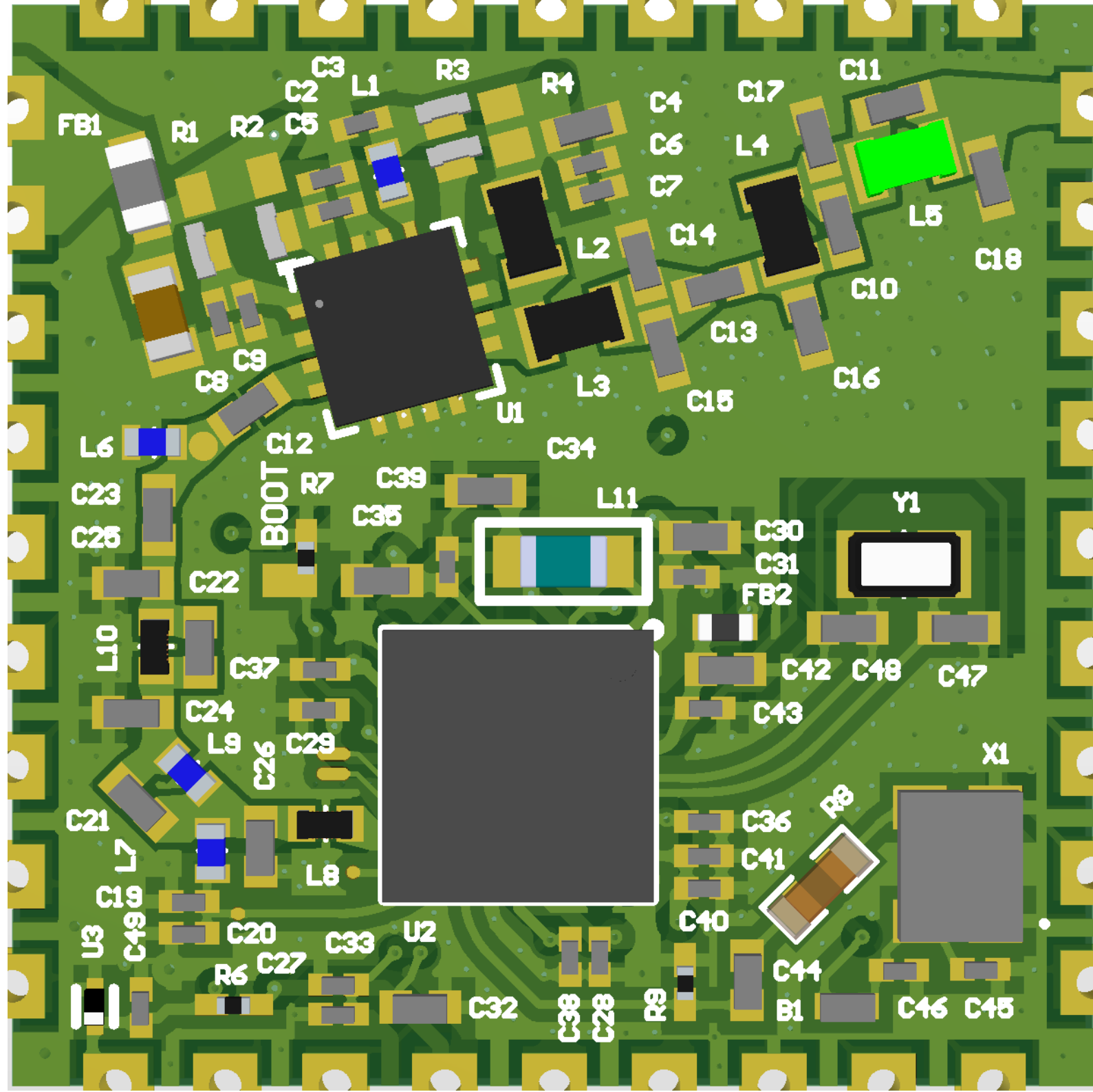


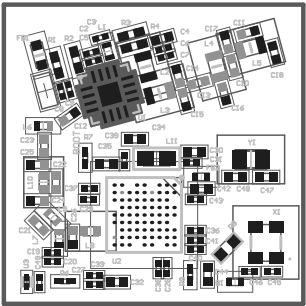
RF power part output (not managed here)





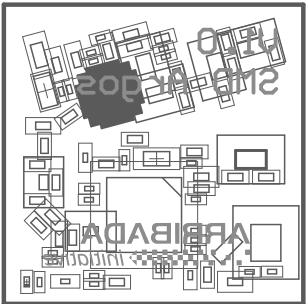
Title <i>HorizonLink-SMD</i>			Arribada Initiative https://github.com/arribadacontact@arribada.org arribada.org UK - Portsmouth	
Size: A4	Reference: argos-sm	Revision: 1.0		
Date: 22/11/2024	Time: 16:59:54	Sheet 5 of 5		
File: D:\Altium\Projects\stm32wl-dev\hardware-lib\DeviceSheets\Power\STM32RF_power\SMPS.SchDoc				





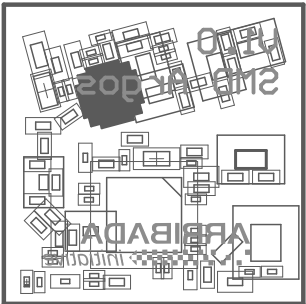
Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0,020mm	3.5	
1	Top Layer		0,035mm		
	Dielectric 1	PP-010	0,071mm	4	
2	Signal Layer 1		0,035mm		
	Dielectric 2	Core-002	1,265mm	4	
3	Signal Layer 2		0,035mm		
	Dielectric 3	PP-010	0,071mm	4	
4	Bottom Layer		0,035mm		
	Bottom Solder	Solder Resist	0,020mm	3.5	
	Bottom Overlay				

Total board thickness: 1,586mm



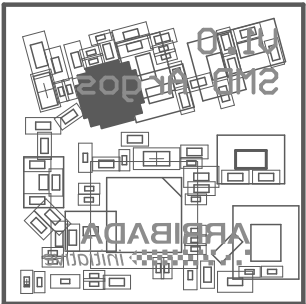
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	Bottom Solder	Solder Resist	0,020mm	3.5	
	Bottom Overlay				

Total board thickness: 1,586mm

Line #	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	1000u@900MHz	Ferrite Bead - BLM15_Z1D Series EMI Suppression Filter 1000 +/- 25% at 100MHz 250mA @85	B1	1	Murata	BLM15HG102S2LD	Volume Production				
	2.2uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.2 uF 10 VDC 10% 0603 X7R	C1	1	Murata	GRM188R71A225KE15D	Volume Production	Mouser	81-GRM188R71A225KE15		
	100nF	Multilayer Ceramic Capacitors MLCC - SMD/SMT .1uF 16V 10% 0201	C2, C3, C6, C8, C19, C31, C33, C36, C37, C38, C39, C41, C43, C46, C49	15	Murata	GRM033Z71C104KE4D	Volume Production	Mouser	81-GRM033Z71C104KE4D		
	1uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1.0uF 10V 10% 0402	C4, C32	2	Murata	GRM155Z71A105KE1J	Volume Production	Mouser	81-GRM155Z71A105KE1J		
	100pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 100PF 25V 5% 0201	C5, C7, C9, C20	4	Murata	GRM035C1E101JX01D	Volume Production	Mouser	81-GRM035C1E101JX1D		
	GRM1555C1H4R3BA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 4.3 pF 50 VDC 0.1 pF 0402 COG (NP0)	C10, C21	2	Murata	GRM1555C1H4R3BA01D		Digi-Key	490-6243-1-ND		
	GRM1555C1E3R9BA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 3.9 pF 25 VDC 0.1 pF 0402 COG (NP0)	C11	1	Murata	GRM1555C1E3R9BA01D		Mouser	81-GRM1555C1E3R9BA1D		
	20pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 20 pF 50 VDC 2% 0402 COG (NP0)	C12, C14	2	Murata	GRM1555C1H200GA01D	Not Recommended for New Design	Mouser	81-GRM1555C1H200GA1D		
	GCM1555C1H101JA16D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 100 pF 50 VDC 5% 0402 COG (NP0) AEC-Q200	C13, C23	2	Murata	GCM1555C1H101JA16D		Mouser	81-GCM1555C1H101JA6D		
	GCM1555C1H8R2CA16J	Multilayer Ceramic Capacitors MLCC - SMD/SMT 8.2 pF 50 VDC 0.25 pF 0402 COG (NP0) AEC-Q200	C15, C25	2	Murata	GCM1555C1H8R2CA16J	Volume Production	Mouser	81-GCM1555C1H8R2CA6J		
	CAP-NC-0402	CMS0402 Non Connected	C16, C22	2							
	GRM1555C1E6R8BA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 6.8 pF 25 VDC 0.1 pF 0402 COG (NP0)	C17	1	Murata	GRM1555C1E6R8BA01D		Mouser	81-GRM1555C1E6R8BA1D		
	GRM1555C2A2R7CA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.7 pF 100 VDC 0.25 pF 0402 COG (NP0)	C18	1	Murata	GRM1555C2A2R7CA01D		Mouser	81-GRM1555C2A2R7CA1D		
	GCM1555C1H120GA16D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 12 pF 50 VDC 2% 0402 COG (NP0) AEC-Q200	C24	1	Murata	GCM1555C1H120GA16D	Volume Production	Mouser	81-GCM1555C1H120GA6D		
	GRM1555C1H130GA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 13 pF 50 VDC 2% 0402 COG (NP0)	C26	1	Murata	GRM1555C1H130GA01D	Not Recommended for New Design	Mouser	81-GRM1555C1H130GA1D		
	100pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 100 pF 25 VDC 1% 0201 COG (NP0)	C27	1	Murata	GRM035C1E101FA01J	Volume Production	Digi-Key	GRM035C1E101FA01JND		
	GRM033R61C104KE14D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0201 0.1uF 16volts XSR10%	C28, C29	2	Murata	GRM033R61C104KE14D		Mouser	81-GRM033R61C104KE4D		
	GRM155Z71A225KE44D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.2 uF 10 VDC 10% 0402 X7R	C30, C35, C42	3	Murata	GRM155Z71A225KE44D	Volume Production	Mouser	81-GRM155Z71A225KE4D		
	0.47uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.47uF 10V XSR10% SMD 0402 85°C Paper T/R	C34	1	Taiyo Yuden	LMK105ABX74KV8F		Mouser	963-LMK105ABX74KV8F		
	1uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1uF +/-10% 10V XSR06 0201	C40, C45	2	Samsung	CL03A106K23NSNC	Volume Production				
	10pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 10 pF 50 VDC 1% 0402 COG (NP0)	C44	1	Murata	GRM1555C1H100FB01D	Volume Production				
	5.1pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0402 MLCC NPO 5.1 pF, +/- 0.1pF 50 V T8R GF	C47, C48	2	Walsin Technologies	0402NSR1B500CT		Mouser	791-0402NSR1B500CT		
	BLM15PE800SH1D	Ferrite Bead, 80 Ohm, 25%, 0402 RoHS Compliant: Yes	FB1	1	Murata	BLM15PB000SH1D	Volume Production	Digi-Key	490-BLM15PB000SH1DCT-ND		
	600R@100MHz	Ferrite Bead 600ohm 500mA 340mOhm 0402	FB2	1	Murata	BLM15B601S2LD	Volume Production				
	9POS_CAST	9 POS Castellated connector with V/A 0.6	J1, J2, J3, J4	4							
	10nH	Inductor RF Chip Multi-Layer 10nH 5% 100MHz 8Q-Factor Air 300mA 260mOhm DCR 0402 Paper T/R	L1, L6, L9	3	Murata	LQG15HS10NJ02D	Volume Production	Arrow Electronics	LQG15HS10NJ02D		
	24nH	Inductor RF Chip Unshielded Wirewound 24nH 2% 100MHz 40Q-Factor Ferrite 1.3A 0603 Paper T/R	L2	1	Murata	LQW18AN24NG80D	Volume Production	Mouser	81-LQW18AN24NG80D		
	LQW18AN5N6C80D	Inductor Wirewound, 5.6 nH, 40 Milliohm, ± 0.2nH, 6.65 GHz, 1.9 A, 0603 [1608 Metric]	L3	1	Murata	LQW18AN5N6C80D	Volume Production	Mouser	81-LQW18AN5N6C80D		
	10nH	Inductor Wirewound, 10 nH, 52 Milliohm, ± 2%, 4.75 GHz, 1.6 A, 0603 [1608 Metric]	L4, L5	2	Murata	LQW18AN10NG80D	Volume Production	Digi-Key	490-17899-1-ND		
	LQG15HN68NJ02D	Inductor RF Chip Multi-Layer 68nH 5% 100MHz 8Q-Factor Air 180mA 1.25Ohm DCR 0402 Paper T/R	L7	1	Murata	LQG15HN68NJ02D	Not Recommended for New Design	Digi-Key	490-1095-1-ND		
	LQG15WZ11NJ02D	Inductor RF Chip Multi-Layer 11nH 5% 0402 Paper T/R	L8	1	Murata	LQG15WZ11NJ02D	Volume Production	Digi-Key	490-15246-1-ND		
	LQG15W120NJ02D	Inductor RF Chip, Aec-Q200, 20Nh, 0402, 2.1GHz RoHS Compliant: Yes	L10	1	Murata	LQG15W120NJ02D	Volume Production	Mouser	81-LQG15W120NJ02D		
	MLZ1608M150W7000	Inductor RF Shielded 15uH, ±5%	L11	1	TDK	MLZ1608M150W7000	Volume Production	Mouser	810-MLZ1608M150W7000		
	2.4K	Thick Film Resistors - SMD 2.4 kOhms 62.5 mW 0402 1%	R1	1	Yageo	RC0402FR132K4L		Mouser	603-RC0402FR132K4L		
	5.42K	Thin Film Resistors - SMD 5.42kOhm 0402 1% 50ppm 63mW/50V	R2	1	KOA Speer	RT73H4LETP5421F50		Mouser	660-RT73H4LETP5421F50		
	0R	Resistors Thick Film - SMD 0402 0.063 W 0hm Jumper	R3, R4	2	Vishay	CRC04020000020ED		Digi-Key	541-0.0JCT-ND		
	100K	Resistors Thick Film - SMD 1/20watt 100kohms 1% 200ppm	R6, R7	2	Vishay	CRC02021100K-FED	Volume Production				
	220R	Resistors Thick Film - SMD HP02 (0402) 1/10W 1% 220R T/R 10000	R8	1	Royal Electronic Factory	HP02WAF2200TCE		Mouser	303-HP02WAF2200TCE		
	10K	Resistor Thick Film Anti-Sulfurated Chip 0201 10kOhm 1% Paper T/R	R9	1	Yageo	AF0201FR0710KL		Mouser	603-AF0201FR0710KL		
	GRF5504	RF Amplifier High Efficiency, 3.5Watt P/A Saturated PA, OP1dB>34.0 dBm; 0.4 - 0.5 GHz	U1	1	Guerilla RF	GRF5504		Mouser	459-GRF5504		
	STM32WL55JC17_de tails	Multi-protocol Wireless 32-bit MCU with LPWAN Sub-GHz radio solution (BGA)	U2	1	STMicroelectronics	STM32WL55JC17		Mouser	511-STM32WL55JC17		
	ESD_6V	ESD protection 6V1 - Varistors MLV0603 18V 3pF	U3	1	STMicroelectronics	ESDALC6V1-U2	Volume Production	Mouser	511-ESDALC6V1-U2		
	TG2520S-MN_32.00-MCGNNM0	TCXO Oscillators TG2520S/MN 32.0000M-MCGNNM0: TCXO 2.8V 3.3V 0.5PPM -40 85C NON OVE	X1	1	Epson	TG2520S/MN32.0000M-MCGNNM0		Mouser	732-2520S/MN32MCGNNM0		
	NX2012SA-32.768K-STD-MUB-1	Crystals 32.768KHz 12.5PF SMD	Y1	1	NDK	NX2012SA-32.768K-STD-MUB-1	Volume Production	Mouser	344-NX2012SA32768K1		